

CSE 3300/5299 Homework 1 (Due on 02/13/2025 Midnight)

Total 100pts (8 questions)

Please submit your answers to all questions in a pdf file.

Note: Please start **Question 8** as early as possible, as installing Mininet may take additional time on certain Mac systems.

Question 1 (5 pts)

A student at UConn is working on a collaborative project with a team in Washington State University (WSU). He has 100 Gigabyte data, and his professor asked him to send the data to WSU as fast as possible. The student has two options, either to send it with a friend who will be traveling to WSU in one week (7 days), or use 1 Mbps dedicated link to transfer the data, which way would be faster (ignore propagation delay)? Explain why.

$$\begin{aligned} 100 \text{ gigabytes} &= 10^8 \text{ bytes} = 8 \cdot 10^{11} \text{ bits} & \frac{8 \cdot 10^{11} \text{ bits}}{1 \cdot 10^6 \text{ bits/s}} &= 8 \cdot 10^5 \text{ seconds} \\ 100 \text{ megabits} &= 1 \cdot 10^8 \text{ bits} \\ 8 \cdot 10^5 \text{ seconds} &\cdot \frac{1 \text{ day}}{8.64 \cdot 10^4 \text{ seconds}} &\approx 9.3 \text{ days} \\ 9.3 \text{ days} &> 7 \text{ days} \text{ so sending it via a friend would be} \\ &\text{quicker since it would take 9 days to transfer it} \end{aligned}$$

Question 2 (20 pts)

Suppose there is a 2 Mbps link shared by multiple users. Each user requires 200 kbps when transmitting, but each user transmits only 20 percent of the time.

- a. How many users can be supported when circuit switching is used, show your steps? (5 points)

$$200 \text{ kbps} = 0.2 \text{ mbps} \quad \frac{2 \text{ mbps}}{0.2 \text{ mbps}} = 10 \text{ users}$$

- b. Assume packet switching is used. Suppose there are 30 users. Find the probability that at any given time, exactly 10 users are transmitting simultaneously. (Hint: Use the binomial distribution. Show the formula and the final calculation result) (5 points)

$$\binom{30}{10} 0.2^{10} \cdot 0.8^{20} = \frac{30!}{10! 20!} \cdot 0.2^{10} \cdot 0.8^{20} \approx 0.0355 = 3.55 \%$$

- c. This is a follow-up question of b. Find the probability that there are at least k users transmitting simultaneously (just list the formula with the parameter k (k can be any integer between 0 and 30)). (10 points)

$$\binom{30}{k} 0.2^k \cdot 0.8^{30-k} = \frac{30!}{k! (30-k)!} \cdot 0.2^k \cdot 0.8^{30-k}$$

Question 3 (10 pts)

Compare the two application architectures: client-server and peer-to-peer, show their advantages and limitations respectively.

A client-server architecture is more hierarchal, with clients specifically pinging a server, while peer-to-peer is not hierarchal and each peer can act as a client and a server. Client-server is very reliable for smaller operations while peer-to-peer is more reliable for larger operations. If there are few clients, one dedicated server is great for managing services, but client-server will slow down if there are too many clients. On the other hand, peer-to-peer works very well for a large amount of clients since the load gets spread out, but if there are few clients then there are few 'servers' and services would not work well since there is no one to take the load. Client-server is also easier to secure since it all happens in one place, and data management is easier since all data is processed in one place. Meanwhile, peer-to-peer can be more resistant to bugs, since it would only bring down one 'server' not the whole system.

Question 4 (5 pts): True or False

- a. A user requests a Web page that consists of a base file and three images. For this page, the client will send one request message and receive four response messages (Consider nonpersistent connections). **False, need one request per response.**
- b. With nonpersistent connections between browser and origin server, it is possible for a single TCP segment to carry two distinct HTTP request messages. **False, nonpersistent can only carry one.**

Question 5 (5pts)

Why is it impractical to eliminate the mail server and instead allow your User Agent to send an email message directly to a recipient?

One big reason is that there would no longer be infrastructure for security and to handle errors. This would make emailing easier to exploit and unreliable. This would also open the issue of handling multiple emails to a recipient at the same time. A mail server could process multiple emails at the same time, but cutting the server out could make it hard on the recipient to process such situations.

Question 6 (20 pts) DNS name resolution

This problem focuses on using DNS to resolve the address of *www.amazon.com* (Fig.1). Answer the following questions.

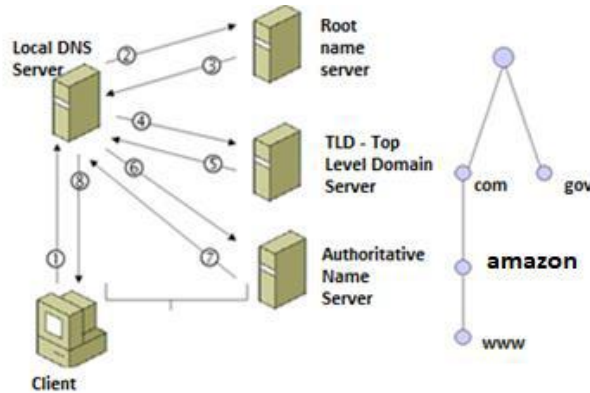


Fig.1 DNS name resolution

- Why might a local DNS server be able to skip sending a query to the Root DNS server? (5 pts)
It may have stored the results from a previous lookup in its cache, and then it can just provide that result again when queried that specifically.
- What type of response is being issued in Step 5, referral or final answer? How do you know? (5 pts)
It is a referral because TLDs do not contain IP address information, which is what the final answer will be.
- What TLD server is responsible for Amazon? (3 pts)
.com
- What type of response is being issued in Step 8? How do you know? (5 pts)
The final answer is being returned. The Authoritative Name Server returns the IP address, which is the last step.
- What is being returned in Step 8? (2pts)
the IP address

Question 7 (20 pts) Delay to load web pages.

Suppose within your web browser you click on a link to obtain a web page. The IP address for the URL is not cached in your client, so a DNS lookup is required. Four DNS servers are visited before your host receives the IP address from DNS: the first DNS server visited is the local DNS cache, with an RTT delay $RTT_0=5$ msecs; the second, third and fourth DNS servers contacted have RTTs of 40, 24, and 50 msecs, respectively. The RTT between the Client and a CNN Web server is $RTT_{\text{HTTP}}=70$ msecs.

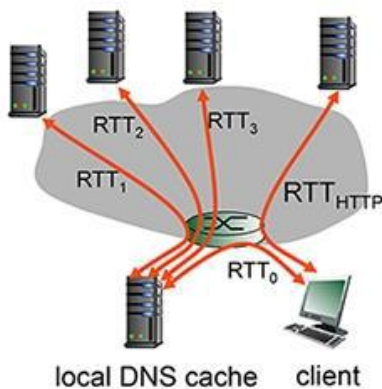


Fig.2 The load of web pages

a. Why do you think the RTT to the local name server is smaller than the other RTTs? (5 pts)

It is smaller because it is closer to the client, meaning it is much easier to reach through the internet than other RTTs.

b. How long does it take for the name *www.cnn.com* to be resolved (assume nothing is cached)? You must show all your work! (5 pts)

$$RTT_0 + RTT_1 + RTT_2 + RTT_3 = 5 + 40 + 24 + 50 = 119 \text{ ms}$$

$$\text{total} = \text{request} + \text{response} = 119 \text{ ms} + 119 \text{ ms} = 238 \text{ ms}$$

c. After the IP address is returned to the client, what 2 steps must be taken before the base HTML web page can be displayed on the browser? (5 pts)

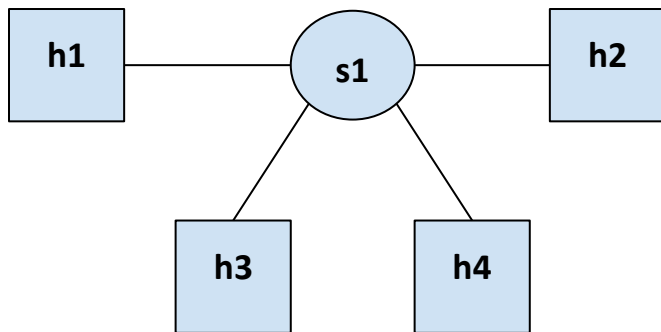
A TCP connection has to be established and the HTML page has to be sent to the client.

d. What is the total delay between clicking on the link and displaying the base web page (including full DNS query in b. and steps in c.)? You must show your work. (5 pts)

$$\text{part b} + \text{TCP connection} + \text{HTML page} = 238 + 70 + 70 = 378 \text{ ms}$$

Question 8 (15 pts): We will use Mininet for project 3. Please follow the provided instructions (uploaded on huskyCT) to install Mininet and answer the questions outlined in the instruction document. The questions are also copied here.

In Mininet change the default configuration to have 4 hosts connected to a switch.



1. Show your screenshot of *dump* and *pingall* output. (5pts). **I included the command I used to set up the four hosts**

```

karinajadia@Ubuntu-2004:~$ sudo mn --topo single,4
*** Creating network
*** Adding controller
*** Adding hosts:
h1 h2 h3 h4
*** Adding switches:
s1
*** Adding links:
(h1, s1) (h2, s1) (h3, s1) (h4, s1)
*** Configuring hosts
h1 h2 h3 h4
*** Starting controller
c0
*** Starting 1 switches
s1 ...
*** Starting CLI:
  
```

```

mininet> dump
<Host h1: h1-eth0:10.0.0.1 pid=2016>
<Host h2: h2-eth0:10.0.0.2 pid=2018>
<Host h3: h3-eth0:10.0.0.3 pid=2020>
<Host h4: h4-eth0:10.0.0.4 pid=2022>
<OVSSwitch s1: lo:127.0.0.1,s1-eth1:None,s1-eth2:None,s1-eth3:None,s1-eth4:None
pid=2027>
<Controller c0: 127.0.0.1:6653 pid=2009>
  
```

```

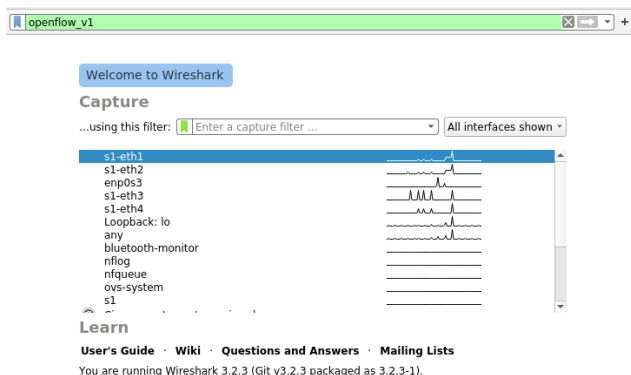
mininet> pingall
*** Ping: testing ping reachability
h1 -> h2 h3 h4
h2 -> h1 h3 h4
h3 -> h1 h2 h4
h4 -> h1 h2 h3
*** Results: 0% dropped (12/12 received)
  
```

2. Run the *iperf* command as well, and screenshot the output, how fast is the connection? (5pts)

```

mininet> h1 iperf -s -p 5201 &
mininet> h2 iperf -c 10.0.0.1 -p 5201 -t 5
-----
Client connecting to 10.0.0.1, TCP port 5201
TCP window size: 6.18 MByte (default)
-----
[ 3] local 10.0.0.2 port 55670 connected with 10.0.0.1 port 5201
[ ID] Interval      Transfer    Bandwidth
[ 3] 0.0- 5.0 sec  15.7 GBytes 26.9 Gbits/sec
  
```

3. Run Wireshark, and using the [display filter](#), filter for “openflow_v1”. Note: When you run Wireshark, you should do so as “sudo wireshark”. When you choose an interface to capture on, you should select “any”. Run ping from a host to any other host using *hX ping -c 5 hY*. Show the screenshot on the Mininet terminal. And take a screenshot to show the captured related packets by Wireshark. (5pts)



```

mininet> h1 ping -c 5 h2
-----
Server listening on TCP port 5201
TCP window size: 85.3 KByte (default)
-----
[ 4] local 10.0.0.1 port 5201 connected with 10.0.0.2 port 55670
[ ID] Interval      Transfer    Bandwidth
[ 4] 0.0- 5.0 sec  15.7 GBytes 26.7 Gbits/sec
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data:
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=1.05 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.411 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.116 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.151 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.101 ms

--- 10.0.0.2 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4061ms
rtt min/avg/max/mdev = 0.101/0.366/1.052/0.360 ms
  
```